Mariam Bouzidi

Tunis, Tunisia — mariambouzidi70@gmail.com — LinkedIn — GitHub

Profile

Data Science Engineer specialized in advanced artificial intelligence systems, with strong academic foundations in machine learning, deep learning, and data-driven system design. Experienced in developing applied AI solutions under industrial constraints and leading interdisciplinary projects, with a strong interest in research at the intersection of artificial intelligence, intelligent systems, and process optimization.

Education

Engineering Degree in Computer Systems (Data Science Specialization) 2022 – 2025
Faculty of Sciences of Tunis (FST), University of Tunis El Manar, Tunisia
Relevant coursework: Artificial Intelligence, Machine Learning, Deep Learning, Probability and Statistics, Data Analysis, Advanced Data Modeling, Big Data Systems, IoT, Data Preprocessing, Business Intelligence, R Programming.

Integrated Preparatory Cycle in Mathematics, Physics, and Computer Science 2020 – 2022
Faculty of Sciences of Tunis (FST), University of Tunis El Manar, Tunisia
Core studies in algebra, analysis, probability, statistics, physics, algorithms, programming, databases, computer architecture, and computer networks.

Experience

Project Manager — AI-Driven Pharmacy Automation System Genove 2025 – Present
Leading the design and development of an AI-based system for pharmacy automation and decision support, coordinating data pipelines, model development, and system integration. Applied machine learning and rule-based approaches to improve operational efficiency in collaboration with domain experts.

End-of-Studies Internship — Data Science Intern Audasoft 02/2025 – 06/2025
Designed and implemented AI modules for an intelligent social media automation platform using open-source LLMs. Built complete data pipelines involving data collection, preprocessing, vector databases, and retrieval mechanisms, with fine-tuning and prompt engineering under constrained computational resources.

Business Intelligence Internship Wevioo 06/2024 – 08/2024
Developed real-time dashboards for biometric card production monitoring by implementing end-to-end data pipelines. Integrated data using Python, Flask, MySQL, and Power BI to enable accurate visualization and decision support.

Machine Learning Internship Smartovate 06/2024 – 08/2024
Built a machine learning system for candidate–position matching using NLP and supervised learning techniques. Conducted data preprocessing, feature engineering, model training, and evaluation to improve matching performance.

Java Developer Internship SOTETEL 06/2023 – 08/2023
Developed a Java-based application for electronic systems security management using MongoDB. Implemented application logic for monitoring, data persistence, and system reliability.

Web Development Internship FCIT 06/2022 – 08/2022
Developed a full-stack web application using Angular, Node.js, and PostgreSQL to manage training and consulting services.

Projects

IoT and AI-Based Process Optimization System
Designed an intelligent system combining simulated sensor data with AI models to monitor and optimize industrial processes. Implemented data preprocessing and efficiency optimization techniques.

Brain Tumor Classification using Deep Learning and Sensor Data
Developed deep learning models for brain tumor classification using multimodal datasets combining medical imaging and sensor-derived data. Performed data preprocessing and model training.

Video Classification Using Multimodal Data
Designed deep learning models combining visual and audio features for video classification. Implemented feature extraction, multimodal fusion strategies, and evaluation pipelines.

Additional Academic Projects

- NLP-Based Conversational Chatbot using machine learning techniques.
- Convolutional Autoencoder for image reconstruction and feature learning.
- Q-Learning algorithm for autonomous pathfinding.
- Machine learning models for metaverse transaction analysis.

Research & Technical Writing

Building AI Without a Budget: Surviving the Open-Source Route (Ongoing)
Technical article focused on designing and deploying AI systems using open-source models and free computational resources under industrial constraints.

Critical Review and Refinement of IoT-Oriented Research Work
Conducted a technical review of an IoT-enabled healthcare research paper, reimplemented the proposed approach, analyzed limitations, and proposed methodological and architectural improvements.

Technical Skills

Programming Languages: Python, Java, C, C++, R, SQL
Artificial Intelligence: Machine Learning, Deep Learning, LLM, Natural Language Processing, Model Fine-Tuning,LangChain, LangGraph
Data Management: MySQL, PostgreSQL, MongoDB, ChromaDB, FAISS
Tools & Platforms: Power BI, Flask, Git, VS Code, Kaggle, Google Colab,Hugging Face, Ollama
Languages: English (C1), French (Fluent), Arabic (Native), Turkish (Conversational), German (Basic)

Extracurricular Activities & Leadership

Founder & Coordinator — Genove (AI-Driven Systems Startup)
Founder & Coordinator — IQCubes AI & Robotics Club
AI Instructor — GoMyCode & CodeQuest (Tunisia)
International Volunteer — AIESEC (Turkey), English Teacher (UN-aligned programs)
Logistics Manager — Google Developer Student Clubs (GDSC)

Certifications

Microsoft AI-900 — Azure AI Fundamentals
Microsoft DP-900 — Azure Data Fundamentals
AWS Academy Cloud Foundations
IBM Machine Learning with Python — Coursera
Microsoft Power BI Data Analyst — Coursera
Generative AI Engineering with LLMs — Coursera Specialization
IELTS